

Bringing the Heat — Hackathon Summary

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A single pipeline covering Levels 0–3 of the brief, run in order:

- 1. **PV-pattern detection** (`src/pv_features.py` , `src/pv_detection.py`) — a `HistGradientBoostingClassifier` on hand-built seasonal/midday-suppression features, validated out-of-fold against the 245 surveyed households (**0.937 ROC AUC, 89.4% accuracy**), then used to label the 165 unsurveyed households so they can join a forecasting group too.
- 2. **Grouped 15-minute aggregation** (`src/aggregate.py`) — PV (158 households), No-PV (252) and all-households (410) consumption series at native 15-min resolution, with weather upsampled from hourly via time-based linear interpolation. The target is each group's **per-household average**, not its raw sum, which is far more stable across the meter-rollout ramp-up and recovers 1–2 extra years of usable history.
- 3. **Day-ahead forecasting** (`src/forecast.py`) — one `LGBMRegressor` per group, predicting all 96 15-minute steps of the next day from a single, realistic **11:45 AM gate-closure cutoff** the day before delivery (never midnight, never same-day weather actuals). Plus two `objective="quantile"` models per group for a nominal 90% prediction interval (Level 3).
- 4. **Level 2 check: did grouping actually help?** — PV and No-PV predictions rescaled back to kWh totals and summed per 15-min step (not by naively adding each group's own MAE, which overstates the error), then compared against one ungrouped model on the same test window.

Headline results (shared test window, 2023-09-05 → 2024-02-27)

Segment	MAPE
PV group	21.6%
No-PV group	11.8%
All-households, ungrouped	12.9%
Grouped, portfolio-wide	12.6%

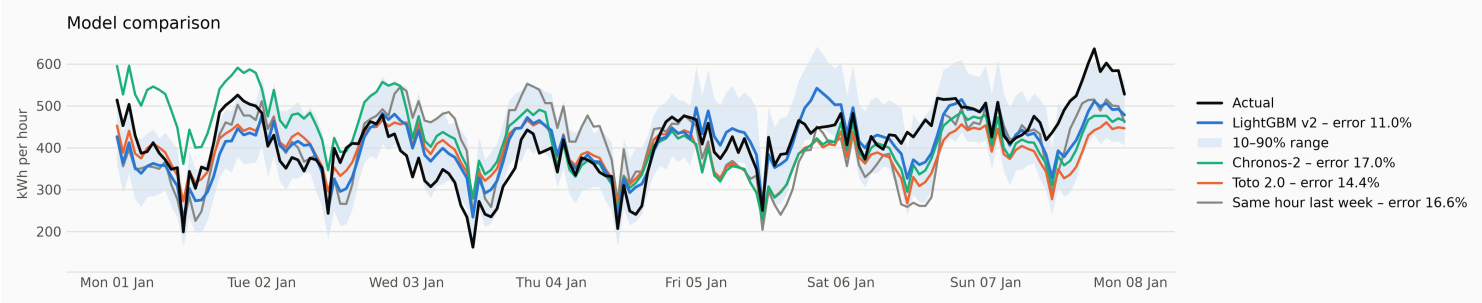
Grouping beats the single ungrouped model on every metric (MAE 16.45 vs 16.64 kWh/15min, MAPE 12.6% vs 12.9%), and the PV/No-PV split exposes an 9.8-point MAPE gap the pooled model hides entirely — useful for procurement risk management on its own. On **uncertainty**, the 90% prediction intervals are honestly overconfident (realised coverage, PICP, is 74–85% against a 90% nominal target, worst for the noisier PV group) — reported as a real calibration gap, not tuned away.

Previous evaluation of different models (full evaluation in `leaderboard.csv`)

Before settling on the LightGBM approach above, a wider model sweep was run across tree, deep-learning, foundation (zero-shot) and naive-baseline families (15 models in total, informative selection shown), scored on portfolio-level normalized MAE (`nMAE`):

Model	Family	Portfolio error (nMAE)	Coverage 10–90 (target 80%)
LightGBM	Tree model	14.07%	76%
CatBoost	Tree model	14.12%	77%
Toto 2.0	Pretrained AI model (no training on our data)	14.88%	78%
Chronos-2	Pretrained AI model (no training on our data)	16.95%	76%
Same hour last week	Simple rule (baseline)	20.42%	77%

Takeaway: gradient-boosted trees (LightGBM/CatBoost, with weather + calendar features, track "W0") win outright — foundation zero-shot models (Chronos-2, Toto 2.0) and trained deep nets (NHITS, TFT, PatchTST) all trail, and every naive baseline is worse still. This is consistent with (and part of the reason behind) the LightGBM/ CatBoost choice used in the pipeline above. `week_strict_chronos_compact.png` below shows one illustrative week of this comparison:



LightGBM (blue, 11.0% error) tracks the actual portfolio load (black) markedly closer than the zero-shot foundation models or the naive same-hour-last-week baseline (grey, 16.6% error), especially around the sharp weekday troughs — the same persistence-style signal the feature-ablation study in `reports/report.md` identifies as the dominant driver of model accuracy.

Outlook: household-level model, evaluated at portfolio level (branch `johannes_12`)

- 1. **Forecast target:** This model predicts the next-day consumption of **each household** rather than that of the total portfolio.
- 2. **Training objective:** This model learns household-level errors; the portfolio model directly minimizes:

$$\min_{\theta} \sum_{(d,h)} (Y_{d,h} - \hat{Y}_{d,h})^2$$

- 3. **Main trade-off:** Household-level modeling provides individual forecasts and flexibility; direct portfolio modeling focuses exclusively on **overall procurement accuracy**. Summed to the portfolio, the household-level model with a squared-error objective reaches **bias +3.1 % and MAPE 8.4 %**.